TFPT — Red Team / Stress Test layer

Five adversarial targets (A–E): try to *break* the reductions, not confirm them

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading document 5 (Red Team).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[E]/[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven   **[C]** conditional   **[O]** open / axiom   **[X]** kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\mathrm{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

What this note is

This is the deliberately *adversarial* layer requested in the v78 review. Its purpose is **not** to confirm TFPT but to attack the five load-bearing reductions at their weakest logical transitions. Each target is run through one fixed protocol — minimal statement, assumptions, logical chain, validity conditions, counterexample search, limiting/degenerate cases, alternative structures, provisional verdict — and produces a deliberately narrow output: where the statement works, where it could fail, and the residual risk. The layer is machine-backed by [`verification/redteam/rt_A_e8net.py`] . . . [`verification/redteam/rt_E_vgeo.py`], [`verification/redteam/rt_F_qft4d.py`] and [`verification/redteam/run_redteam.py`]; a red-team check asserts an *adversarial* fact (a counterexample really exists, a hidden assumption is really needed, a firewall really holds), and the honest outcome lives in the **status** of each target, never in a green pass.

Target	Minimal claim	Status	Residual risk
A	seam–Calderón measure = $(E8)_1$ net	**reduced, not closed**	*one* statement: boundary-net holomorphy + c=8 ($\Leftrightarrow$ the index-4 inclusion); E_8 and bulk uniqueness then automatic (v83/v87/v89; Lie-level realisation v143). Since reduced to *zero new* open content: the free-bulk premise is a fixed-point theorem, the cone is eliminated (cone gap = one-particle gap), and the residual factors into the already-open A_2 (net assembly) + `GATE.QGEO` (v160–v165). The three-residual form below is the historical development.
B	$g_{\mathrm{car}} = 5$ forced by the Pascal rule	**survives (narrowed)**	boundary derivation of half-spinor exhaustion (degree-2 truncation)
C	$k = c_3/2 = \frac{1}{16\pi}$, $S/A = \frac{1}{4}$	**survives**	absolute $1/G$ (UV-sensitive) is the one anchor
D	ratios+products $\Rightarrow$ one scale v_{geo}	**survives (narrowed)**	CP phases ($\delta_{\mathrm{CKM/PMNS}}$) not covered by v_{geo}
E	one scale v_{geo} carries the theory	**survives (narrowed)**	EW/reheating/leptogenesis scales + seam=Planck identification

Method

The red team treats each reduction as a hostile witness. A confirmatory script that always passes is worthless here: the layer is built so that it *may* downgrade a claim. Three outcomes are allowed — **survives** (stands as worded), **survives (narrowed)** (stands only after a silent assumption is made explicit), and **reduced, not closed** (the conservative wording is the correct one). A fourth, **broken**, is reserved for an actual failure; none occurred.

Target A — the $(E8)_1$ boundary-net identification

Minimal claim. The ambient metric/projective measure is reduced to identifying the seam–Calderón boundary measure with the $(E8)_1$ lattice net, carrier subnet $(D5)_1 \times (A3)_1$. [`verification/redteam/rt_A_e8net.py`]

Logical chain (established). The Sugawara level-1 central charges are $c(E8)_1 = \frac{248}{31} = 8$, $c(D5)_1 = 5 = g_{\mathrm{car}}$, $c(A3)_1 = 3 = N_{\mathrm{fam}}$, so the conformal-embedding criterion $c_{\mathrm{coset}} = c(E8) - c(D5) - c(A3) = 0$ holds [E]. This is *compatibility.*

Where it breaks: central charge underdetermines the net

Counterexample: $(D8)_1 = SO(16)_1$ also has $c = \frac{120}{15} = 8$, but is *not* holomorphic (four primaries $1, v, s, c$) — a distinct $c = 8$ net. Hence equal central charge is *necessary, not sufficient*: $(E8)_1$ is the unique *holomorphic* $c = 8$ net (even unimodular rank-8 lattice $= E8$), so **holomorphy (single vacuum sector, μ-index 1) is the load-bearing extra assumption**. Dropping chirality is worse: the $c = 8$ Narain family is positive-dimensional. The constructive map seam–Calderón kernel $\to (E8)_1$ and the uniqueness of bulk reconstruction are not established; the gap $\Delta_{\mathrm{eff}} > 0$ *supports* tightness (clustering) but does not prove it.

Verdict. Keep "reduced to a rigorous boundary-net identification problem"; do *not* write "metric sector closed". The red team confirms the conservative wording. Status [C]. Residual: holomorphy proof + constructive boundary map + bulk-reconstruction uniqueness.

Target B — carrier rank / the Pascal condition

Minimal formal claim. $2^g = g^2 + g + 2$ has the unique solution $g = 5$ (Lean-verified [E]). This arithmetic is *not* attacked. The attack is upstream: why must the carrier obey the Pascal half-spinor exhaustion $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$? [`verification/redteam/rt_B_pascal.py`]

Where it narrows: the rule, not the arithmetic, selects 5

Perturbing the constant breaks it: $2^g = g^2 + g + c$ gives $\{5\}$ only at $c = 2$; neighbours give other or empty solution sets. The truncation *degree* is the real choice: with $2^{g-1} = \sum_{k \le K} \binom{g}{k}$ one finds $K{=}0 \to g{=}1$, $K{=}1 \to 3$, $K{=}2 \to 5$, $K{=}3 \to 7$, i.e. $g = 2K + 1$. So choosing $K = 2$ (to reach $g_{\mathrm{car}} = 5$) is a physical postulate. It is corroborated — the $D5$ half-spinor $\dim S^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$ and the family closure $N_{\mathrm{fam}} = (2^{g-1}-1)/g$, $g + N_{\mathrm{fam}} = 8$ both pick 5 — but each is itself a postulate, not a theorem.

Verdict. Theorem A's arithmetic core stands; the Pascal *selection* must be typed [O]/[C], not [E]. Residual: a boundary/seam derivation of half-spinor exhaustion.

Target C — $k = c_3/2$ and the seam-area coefficient

Minimal claim. In reduced units $k = c_3/2 = \frac{1}{16\pi}$ and Fursaev–Solodukhin gives $S/A = \frac{1}{4}$. [`verification/redteam/rt_C_kc3.py`]

The dimensional firewall holds

The only legal chain is

$$k_{\rm red} = \tfrac{c_3}{2} = \tfrac{1}{16\pi} \text{ (dimensionless)}, \qquad k_{\rm phys} = \tfrac{k_{\rm red}}{G}, \qquad S = 4\pi\, k_{\rm phys} A_{\rm phys} = \tfrac{A_{\rm phys}}{4G}.$$

The forbidden form $k_{\rm phys} = c_3/2$ is dimensionally false (legal only at $G = 1$). A scan of `v67`/v68/`v73` finds every $c_3/2$ used as the *reduced* coefficient and **zero** naked dimensionful identities; the firewall's teeth are verified against a synthetic violation. Internally consistent with $1/G$ being UV-sensitive (Sakharov/Connes, v68).

Verdict. Safe as worded (reduced coefficient). Status [E] (structure) + [O] for the residual. Residual: the absolute $4D$ Newton scale $1/G$ ($\propto \Lambda^2 f_2$) stays the one dimensionful anchor; nothing *dimensionless* is open.

Target D — $U_{\rm point} \to v_{\rm geo}$

Minimal claim. Ratios plus sector products determine the dimensionless amplitudes, leaving one common scale $v_{\rm geo}$. [`verification/redteam/rt_D_upoint.py`]

Where it narrows: the bijection needs explicit hypotheses

For positive reals, (ratios, product) $\Leftrightarrow$ (individuals) is a bijection, and the TFPT sectors satisfy it (positive rational c, odd size $3 = N_{\rm fam}$). But it *fails* off-assumption: even sector size leaves a sign ambiguity ($\{2, 3\}$ vs $\{-2, -3\}$ share ratio $3/2$ and product 6); complex amplitudes leave an n-th-root branch ambiguity ($\{1, 1, 1\}, \{\omega, \omega, \omega\}, \{\omega^2, \omega^2, \omega^2\}$ share all ratios and product 1); a zero amplitude is degenerate. **Genuine residual:** the bijection fixes amplitude *magnitudes* only — CP phases $\delta_{\rm CKM}, \delta_{\rm PMNS}$ are complex and are *not* folded into $v_{\rm geo}$.

Verdict. Holds for TFPT once four hypotheses (real, positive, fixed branch, no zeros / odd sector size) are made explicit; they are silent in v75 and must be named. Status [E] (reduction) + [O] ($v_{\rm geo}$). Residual: the CP-phase sector.

Target E — $v_{\rm geo}$ as the unique dimensional floor

Minimal claim. No pure-number theory can derive an absolute dimensionful scale; all scales are ratios to one unit $v_{\rm geo}$. [`verification/redteam/rt_E_vgeo.py`]

Single-scale audit: exact for the certified tiers, conditional beyond

The closed compiler + protected IR readouts all reduce to (pure number)$\times v_{\rm geo}$: $M_{\rm scal} = c_3^{7/2} \bar{M}_{\rm Pl}$, $f_a = (c_3/|\mu_4|)^{7/2} \bar{M}_{\rm Pl}$, masses $= (\pi/\sqrt{2})\, c_f\, \varphi_0^k\, v_{\rm geo}$, and $1/G$ shares the same anchor (v68/v75). The audit *surfaces* candidates that are not pure reductions: the seam cutoff Λ (only an *identification* seam=Planck collapses it to $v_{\rm geo}$), the electroweak matching scale, the reheating temperature, and the leptogenesis scale — all frontier [C]/[O] inputs that must not be silently absorbed into $v_{\rm geo}$.

Verdict. $v_{\rm geo}$ is the unique dimensional floor of the certified tiers (true by dimensional analysis). Full single-scale-ness is conditional on the flagged identifications staying honestly typed. Status [E] (ratios) + [O] (one scale). Residual: explicit reduction (or honest frontier typing) of the EW / reheating / leptogenesis scales + the seam=Planck cutoff.

Target F — the perturbative 4D-QFT + scale round (v269–v275)

Minimal claim. The round published in release 5.2 — the Epstein–Glaser $S_{\rm pert}$ layer (v269/v271/v273), the PMNS Jarlskog CP strength (v270), the absolute ν-scale typing (v272), the scale over-determination (v274) and the `QG.AMB.01` roadmap (v275). [`verification/redteam/rt_F_qft4d.py`]

Two attacks land; the round survives once they are named

(1) The gravity S_{pert} is non-unitary. Power-counting renormalizability is *not* unitarity: the R^2/Weyl^2 sector gives a four-derivative propagator $1/(p^2(p^2+M^2))$ with a negative-residue massive spin-2 mode (the Stelle ghost). So S_{pert} is unitary for the matter+gauge sector only; the gravity perturbative S-matrix carries the ghost — which is exactly why the *nonperturbative* `QG.AMB.01` is the real frontier, not a new gap.
(2) The over-determination is conditional. v274's Route-2 inverts the $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ prediction, which is itself `LAMBDA.BRANCH.01` [C]; so the 0.11% gravity-vs-cosmology agreement is conditional on the Λ-branch (the gravity route is unconditional). **Already-honest:** J_{PMNS} inherits δ's [C] provenance (assembled μ_6 phase, not an M_ν output) and the ν-scale is [O] with the NO floor $\Sigma m_\nu = 0.0586$ eV as the one kill test.

Verdict. Survives (narrowed): the round stands as typed; the two narrowings above are now folded into v269 and v274. Residual: the perturbative gravity S-matrix is non-unitary (the Stelle ghost) — this *is* the `QG.AMB.01` frontier; and the anchor over-determination is conditional on the Λ-branch.

Outcome

What the red team changed. No target broke. The hardest gate (A) is confirmed *open* with the conservative wording; four targets (B, D, E, F) *survive once a silent assumption is named*; one (C) survives as worded. The net effect is not new physics but sharper typing: the package now states *where* each reduction would fail and *which* assumptions are truly necessary, exactly at the weakest logical transitions. Reproduce with `cd verification/redteam && python run_redteam.py`.

Follow-up — the verdicts were then used as a worklist [E] ([`verification/v83_e8net_holomorphic_uniqueness.py`])

Two residuals were attacked directly after the red-team pass; the result is recorded in ledger rows `GATE.METRIC.04` and `CAR.PASCAL.01`.

Target A, residual 1 — closed [E]. At level 1 the number of primaries equals det(Cartan) = |fundamental group|: A_3=4, D_5=4, D_8=4, $\mathbf{E_8}$=**1**. So holomorphy (single vacuum sector, μ-index 1) is *necessary* — it excludes the same-c rival $(D_8)_1 = SO(16)_1$ ($c = 120/15 = 8$ but four primaries $1, v, s, c$) — *and sufficient*: a holomorphic $c = 8$ chiral CFT is the lattice theory of an even unimodular rank-8 lattice, of which there is exactly one, E_8, by the Minkowski–Siegel mass formula (mass $= 1/|W(E_8)| = 1/696729600$, and $|\text{Aut}(E_8)|$ saturates it). The $(D_5)_1 \times (A_3)_1$ embedding $c = 5+3 = 8$ stays *compatibility* (16 primaries vs 1). **Consequence:** the constructive map need only show the seam–Calderón boundary net is holomorphic with $c = 8$ (then E_8 is automatic), so Target A drops from *three* residuals to *two*: (i) boundary-net holomorphy $+c = 8$ ($\Delta_{\text{eff}} > 0$ supports, not proves), and (ii) bulk-reconstruction uniqueness — both still [C]/[O].

Target B — reduced [E]. The "$K = 2$ Pascal truncation" is not free: $K = (g-1)/2$ is the Pascal-row midpoint $\sum_{k \le (g-1)/2} \binom{g}{k} = 2^{g-1}$ (odd g), i.e. the half-spinor split; the carrier is the even Clifford half-spinor $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$. So B's residual reduces to the single standard input "carrier = half-spinor of Spin(10)" (the Lean arithmetic core stays [E]).

Target B — a backward certificate ([`verification/v219_icosahedral_mckay.py`]). The choice $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}) = (2, 3, 5)$ that B attacks is, *given* the E_8 closure, the icosahedron: by the McKay correspondence E_8 is the exceptional top of the finite-$SU(2)$ tower ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$), the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ equal to the affine-E_8 Kac marks ($\sum = 30 = h(E_8)$), and it contains the $2, 3, 5$ rotation-axis orders. This does *not* break B's verdict — it is a *backward* certificate (it presupposes E_8), so $g_{\text{car}} = 5$ stays the P2 interface [O]; what it removes is the impression that $(2, 3, 5)$ is an arbitrary triple, since no spherical McKay node sits above the icosahedron.

Still open, typed honestly: A (i)+(ii); Target D's CP-phase residual (at $\theta_{13} = 0$ the Dirac phase is undefined, so this needs a texture correction); Target E's reheating/leptogenesis scales and the seam=Planck identification. Target C is a dimensional *floor*, not a gap. **No target was promoted.**

Second follow-up round [E] ([verification/v87_bulk_uniqueness_reduction.py], [verification/v88_cp_phase_audit.py], [verification/v86_nstar_reheating.py])

Target A, residual (ii) — conditionally closed [E]: Target A = ONE residual. Bulk-reconstruction uniqueness is *not* independent of residual (i). By the algebraic-QFT classification of full 2D CFTs over a chiral net (Longo–Rehren; Kawahigashi–Longo–Müger; Bischoff–Kawahigashi–Longo–Rehren), the possible bulks are the haploid commutative Q-systems $\simeq$ physical modular invariants of $\mathrm{Rep}(A)$. For a *holomorphic* net $\mathrm{Rep}(A) = \mathrm{Vect}$, so the bulk pairing is *unique*. The contrast is machine-checked: the same-c rival $SO(16)_1$ (discriminant category $\mathbb{Z}_2\times\mathbb{Z}_2$, μ-index 4) admits *six* modular invariants — diagonal, $s\leftrightarrow c$ swap, both E_8-extension invariants $|\chi_0+\chi_{s/c}|^2$ (the lattice glue $E_8 = D_8 \cup (D_8+s)$, h_s=1 bosonic), and two twisted pairings — so without holomorphy the $c = 8$ bulk is *multiply* ambiguous; with it, trivially unique. Hence the entire Target A reduces to the single theorem "seam–Calderón boundary net is holomorphic with $c = 8$" ([C]/[O]; ledger `GATE.METRIC.05`). The theorem also has an equivalent, more tractable *index* form ([verification/v89_carrier_index_lemma.py], ledger `GATE.METRIC.06`): by Kawahigashi–Longo–Müger, $\mu_A = [B{:}A]^2\mu_B$, the carrier inclusion has Jones index $[(E_8)_1 : (D_5)_1{\times}(A_3)_1] = \sqrt{16} = 4 = |\mu_4|$ — the glue-group order *is* the inclusion index — and the extension is the μ_4 simple-current extension (all three glue sectors are $h = 1$ currents; $248 = 45{+}15{+}64{+}64{+}60$). Since an index-4 simple-current extension of the carrier has $\mu = 16/4^2 = 1$, **holomorphy follows from the index statement**: Target A $\Leftrightarrow$ "the seam net contains the carrier net with Jones index 4 as its μ_4 simple-current extension" — both ends constructed objects, the index controllable by Longo theory, the gap supporting finiteness. And *which* extension carries no freedom ([verification/v92_glue_uniqueness.py], ledger `GATE.METRIC.07`): exhaustive classification of the carrier discriminant form ($\mathbb{Z}_4{\times}\mathbb{Z}_4$, $q{=}(5x^2{+}3y^2)/8$) shows the extension tower is *completely rigid* — exactly two Lagrangian glues (the two chiralities, identified by the sheet $\mathbb{Z}_2$; the same two objects as the E_8-extension invariants above), and exactly one halfway extension, whose induced form $q = \{0, 0, 0, \frac{1}{2}\}$ *is* the D_8 discriminant form: the $SO(16)_1$ rival is the unique intermediate of the tower, not an arbitrary counter-model. Tower: carrier (μ=16) $\to$ $SO(16)_1$ (μ=4) $\to$ $(E_8)_1$ (μ=1, sheet pair) — nothing else exists. Target A is thereby the *bare* index statement.

Target A, final form — the Simple-Current Extension Theorem [E]/[C] ([verification/v154_simple_current_theorem.py]). Assembled into one statement: the carrier net $A = (D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L = \langle(1, 1)\rangle$ has $[B : A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$. The *entire* adversarial residual of Target A is now the single seam-side identification "the RP seam–Calderón net *is* A" — the algebra ($B \cong (E_8)_1$) is exact and machine-checked (v89/v92/v125/v143/v148). That residual is now sharply located: the analytic core ("the kernel is quasi-free") *follows* from a free RP bulk — the boundary marginal of a Gaussian bulk is the compression of a contraction (v155), the *same* premise the area law and the EH mechanism use — and the net is *explicitly constructed and verified*: DtN = $|k|$, 16 free Majoranas, currents 248=120+128, holomorphic character $E_4/\eta^8 = j^{1/3}$ with 248 at level 1 (v156). *Final reduction (v160–v165):* the free-bulk premise is recast as a fixed-point theorem (quasi-free $\Rightarrow$ all higher cumulants κ_{2n}=0, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), whose value is forced by μ_4-equivariance (v162); and the remaining 16-dim identification factors into the two *already-open* items — A_2 net existence (an assembly of established net constructions, [C]) and R_1 = `GATE.QGEO` ("seam boundary = $\mathbb{P}^1\backslash\mu_4$") — with the irreducible core $\{\pi, v_{\rm geo}\}$ a theorem (v165). So Target A carries *no new independent open content*: it is a unification, not an unconditional proof. *The two structural residuals are then discharged to a theorem (v175):* the CAR second-quantisation functor $\Gamma(t){=}\bigoplus_m \Lambda^m(t)$ makes full-cone reflection positivity reduce *for every m* to the one-particle contraction (verified on the complete 2^{16}-dim Fock space: contraction, fixed multiplicity 2^8, sub-leading = $(2/3)^6$), and A_2 is an assembled and verified $(E_8)_1$ certificate (character $1{+}248q{+}4124q^2{+}\dots$, embedding 248=120+128, E_8 Cartan even unimodular). Both then need the *same* single input — the seam realisation `QGEO.REALIZE.01` — which stays the one irreducible [O]; net existence and full-cone RP become [E]. *That single residual is then driven to bedrock (v176–v181):* it is assembled as one central theorem (v176), split into the marks + kernel obligations whose finite cores close (v177/v178), unified into one conformal-realisation premise (v179), and reduced — via uniformisation, Kerékjártó and the order-4 Möbius classification — to a seam-*isometry* premise (v180) and finally to the strictly weaker conformal-*symmetry* / deck premise `QGEO.SYM.01` (v181): the carrier μ_4 clock is the conformal deck of the seam. Below that the residual is *definitional*, not a missing theorem — the single irreducible structural residual of the whole theory. It is since driven to its sharpest, *non-circular* form (v194–v198): the raw seam DtN is RP-canonical

(Osterwalder–Schrader on the quasi-free state), and the bedrock is cracked to the *state-invariance* $\omega \circ \rho = \omega$ — the DtN principal symbol commutes *exactly* with the clock, and Tomita–Takesaki gives $[\rho, \Lambda_\Sigma] = 0$ with no conformal-covariance assumption, removing the Bisognano–Wichmann circularity — the maximally-operational form of the same postulate, still [O]. *The geometric and conditional cores are now Lean-formalised* (AUDIT: PASS): the UNIFORM normal form ($z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit$\to \mu_4$; `FORM.QGEO.02`) and the implication mark-local$\Rightarrow \omega \circ \rho = \omega$ (`FORM.QGEO.01`) — so the deduction below the premise is machine-proved, while `QGEO.SYM.01` itself stays the one [O] postulate.

Target D — the CP-phase residual quantified [E] (not closed). With the frozen leading reading $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$: pull vs $\gamma_{\rm PDG} = 65.7^\circ \pm 3.0^\circ$ is $+0.98\sigma$ (survives). Audit, *not* promoted: the data central value sits 0.07° from the alternative coefficient reading $\pi/3 + 2\lambda_C^2$ ($|\mathbb{Z}_2|$ instead of $N_{\rm fam}$) — a textbook look-elsewhere trap; the registry keeps coefficient 3 (`REG.FREEZE.01`). Decision threshold: the readings differ by $\lambda_C^2 = 2.88^\circ$, i.e. 3σ-distinguishable once $\sigma_\gamma \le 0.96^\circ$. The J-inversion is flagged magnitude-contaminated (source-scale s_{23} vs M_Z). Ledger `FLAV.CP.01`.

Target E — one flagged scale computed [C]. The reheating temperature is no longer a silent input: $M_{\rm scal} = c_3^{7/2} \bar{M}_{\rm Pl}$ [E] + standard physics give $T_{\rm reh} = 9.6 \times 10^9$ GeV and $N_\star(0.05/{\rm Mpc}) = 51.4$ [C] (ledger `COSMO.NSTAR.01`); honestly recorded: this slow Higgs-channel point is A_s*-disfavoured* (-11.4σ; the measured A_s requires near-instantaneous reheating), so the point is conditional on the decay channel and the frozen band $[50, 60]$ stays the surface of record; the leptogenesis scale and the seam=Planck identification remain typed inputs. **Again: no target was promoted; one residual was *merged* (A), one was *quantified* (D), one input was *computed* (E).**

Target D — the CP residual geometrically *located* (not closed) [E]/[C]

The follow-up round only *quantified* the CP residual; this round *locates* it geometrically, in three consistent readings, without promoting it. **(i) A channel typing ([`verification/v227_degree_exponent_channel_split.py`]).** The E_8 split $248 = 120 + 128$ is a magnitude/phase split: the exponents sum to $120 = |R^+(E_8)|$ (the *magnitude* channel, where the $(\text{ratios}, \text{product}) \Rightarrow v_{\rm geo}$ bijection lives), the fundamental degrees sum to $128 = \operatorname{rank} E_8 \cdot \dim S^+$ (the *phase/glue* channel). The un-covered CP phases of Target D are exactly the 128 phase channel — a clean home, not a new gate; whether 128 also hosts a dark sector is Frontier, *not* asserted (this replaces the over-strong "S^- is dark matter" reading). **(ii) A hexagonal modulus ([`verification/v220_cp_hexagonal_modulus.py`]).** The seam deck is the *square* modulus (j=1728, $\mathbb{Z}/4$ clock; the kill-test selects it), which is real and carries no phase; its exceptional partner is the *hexagonal* modulus j=0 (τ=ρ=$e^{i\pi/3}$, $\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$). The frozen leading reading $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$ has leading term $\pi/3 = \arg \rho$, the hexagonal CM phase; CP lives in the $\mathbb{Z}/6$ phase *fiber* over the $\mathbb{Z}/4$ seam, not in the deck. **(iii) An orientation ([`verification/v225_dual_normal_frame.py`]).** The dual normal frame $d = a^\top R^{-1} = -\frac{1}{2}(1, 1, -2)$ (Nariai/horizon normal) and $n = (5, -9, 6)$ (first-generation selector, $n^\top R = (8, 0, 0)$) has oriented volume $\det(\mathbf{1}, d, n) = 21 = N_{\rm fam} \cdot 7$, and rotating by the hexagonal unit gives $\operatorname{Im} \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3) \neq 0$ — CP sits precisely in the *orientation* of the normal frame, exactly where the magnitude bijection fails. All three are [E] geometry; the physical CP *derivation* stays the Target-D [C] residual (the seam deck is not re-selected, the registry δ is untouched). **(iv) One hexagonal unit, two sheets ([`verification/v231_cp_mu6_phases.py`]).** The reduction is sharper still: *both* CP phases are μ_6 powers of the single hexagonal CM unit $\rho = e^{i\pi/3}$ — $\delta^{\rm lead}_{\rm CKM} = \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\rm PMNS} = \arg(\rho^4) = 4\pi/3$ (lepton, the neutrino phase lattice) — with $\rho^4 = -\rho$, so $\delta_{\rm PMNS} = \delta^{\rm lead}_{\rm CKM} + \pi$ is the quark phase times the $\mathbb{Z}_2$ sheet ($\rho^3 = -1$); the frame orientations of (iii) are correspondingly sheet-flipped ($\pm 21 \sin(\pi/3)$). So Target D's CP sector is *not* two independent phase inputs but *one* hexagonal unit read on the two sheets — the magnitude residual $3\lambda_C^2$ (quark, v88) and the physical derivation stay [C], but one of the two phase inputs is removed. **(v) The universal triality phase ([`verification/v233_cp_triality_phase.py`]).** In the canonical $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$ factorisation, $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight); since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) phases have the *same* family/triality class and differ *only* by the sheet. So the CP phase *is* the universal triality phase, sheet-split — not even a free choice of μ_6 power: the power choice of (iv) is removed, and ω is the μ_3 factor of the order-30 $(2, 3, 5)$ Coxeter/Milnor monodromy that builds the hull (v232). The physical CP derivation still stays [C].

Global look-elsewhere closure [E] ([`verification/v100_numerology_null_mc.py`])

The per-target look-elsewhere traps recorded above (the $\pi/3+2\lambda_C^2$ alternative in Target D, the $\ln(46080)$ scale reading destroyed in [`verification/v99_koide_flow_time.py`]) are local instances of the global adversarial question: *could the whole scorecard be formula-fishing?* The numerology null test answers it with an explicit, declared null model run in adversarial mode: the *entire* complexity-matched formula grammar (which provably contains every scored TFPT formula — the null space includes the theory) is enumerated against conservative data windows. Joint formula-fishing probability $\prod_i p_i = 10^{-25.8}$ over the 13 scored observables; all 94 500 $F_{U(1)}$ variants solved, exactly one hits CODATA ($p_\alpha \leq 1.1{\cdot}10^{-5}$); total $\leq 10^{-30.7}$. The test is falsifiable in both directions: the negative controls (seed perturbation 1%/10% collapses TFPT's own hit count $13 \to 9/6$; data shuffling $\to 1.2$) prove it has power, and the census is stable under budget/window variation (< 2 orders). Honest limit, stated as such everywhere: this is a null-model rejection *conditional on the declared grammar* — no statistical test certifies a theory; the tension observables keep their large windows and contribute nothing to the number. Details and the grammar definition: the audit note (`tfpt_3`) and the module docstring.

External-review residual map [E]/[C] ([`verification/v182_reviewer_residual_map.py`])

A careful external review (2026-06-14) raised four concerns at the [C]/[O] seams: **(A)** the mapping $2D\to4D$ ("why does the dynamics read the masses as the Plücker norm 11?"), **(B)** the abstractness of UV gravity, **(C)** the meta-look-elsewhere of the grammar choice, and **(D)** the Koide Möbius flow (non-integer flow time $t\sim2.84$, missing QFT generator). None adds an *independent* open gap. **A** [E]/[C]: the readout integer is exact ($11 = \sum_{k\leq2}\binom{4}{k} = \dim S^+ - g_{\mathrm{car}}$, the QBL boundary count, v174) and the $2D\to4D$ step is the holographic reduction, so $A = \texttt{QGEO.SYM.01} + F_{\mathrm{transfer}}$. **B** [E]: "gravity is the geometric readout of the boundary" *is* `QGEO.SYM.01`. **D** [E]/[C]: $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2,\mathbb{C})$ is the Möbius group and the cusp flow lives on the deck $\mathbb{P}\setminus\mu_4$, so the Möbius *nature* is forced by the geometry; the non-integer $t\sim2.84$ only locates the physical point and the missing generator *is* F_{transfer}, so $D = \texttt{QGEO.SYM.01} + F_{\mathrm{transfer}}$. So A, B, D collapse onto the *two already-named* residuals; only **C** is genuinely *experiment-only*: the grammar is frozen (v84), minimal and over-determined (the anchor $a = (1,1,2)$ lands in ≥ 6 mutually-independent, not-selected-for structures), a look-elsewhere-resistant signature whose residual — correctly — falls only to out-of-sample data (JUNO, CMB-S4), never to argument. The four concerns are a unification of the residual *structure*, not a closure of the underlying premises, which stay [O]/[C].